

AMPLITUDE-DEPENDENT FIRST SPACING AND THE CIRCLE AND DIVISOR PROBLEMS

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ABSTRACT. We prove that the error terms in Gauss's circle problem and Dirichlet's divisor problem are $O_\varepsilon(X^{\vartheta+\varepsilon})$, where

$$\vartheta = \frac{33333 - 8\sqrt{64515}}{99856} = 0.3134615421015514\dots$$

The principal estimate is a fourth-to- q th moment bound for the discrete cone frequencies $(l, kl, l\sqrt{k})$, uniform in bounded coefficients. It gives a square-root bound for $4 < q \leq 9/2$ whenever $K^{q-4} \leq L$ and $L^{3q-8} \leq K^{8-q}$. In particular, the $q = 9/2$ estimate holds for $L^{11/7} \leq K \leq L^2$. The proof separates the same-ray and cross-ray terms inside the local square function before applying the amplitude-dependent wave-envelope estimate. The arithmetic second-spacing input and block lengths remain those used by Li and Yang.

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1. INTRODUCTION

1.1. **The circle and divisor errors.** For $X \geq 2$, let

$$R(X) = \#\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \leq X\} - \pi X, \tag{1.1}$$

$$\Delta(X) = \sum_{n \leq X} d(n) - X \log X - (2\gamma - 1)X, \tag{1.2}$$

where $d(n)$ is the divisor function and γ is Euler's constant. We use the squared-radius variable X in the circle problem throughout. Thus an exponent θ here corresponds to the exponent 2θ in the radius variable. The conjectural pointwise bound in both problems is $O_\varepsilon(X^{1/4+\varepsilon})$; see, for example, the introduction of Li–Yang [11].

Date: September 6, 2026.

2020 Mathematics Subject Classification. 11P21, 11L07, 11N37, 42B10, 42B25.

Key words and phrases. Gauss circle problem, Dirichlet divisor problem, first spacing, wave envelopes, cone, large values.

The relation between these two problems is more substantial than the similarity of their conjectured exponents. Hyperbola decompositions express both errors in terms of fractional parts of reciprocal functions. After Fourier expansion, one obtains double sums in an integer frequency and an integer position variable. The Bombieri–Iwaniec method [1], adapted to the circle and divisor problems by Iwaniec and Mozzochi [10], divides the position interval into short pieces, classifies their slopes by rational approximations, and applies Poisson summation. A double large sieve then separates the resulting estimate into two spacing problems.

The first spacing problem concerns additive relations among the transformed frequencies. The second concerns coincidences among the rational and curvature data attached to the original short intervals. Huxley’s sequence of works [7, 8, 9] developed resonance-curve methods for the latter problem. In the squared-radius normalization, [9] gives the exponent $131/416 = 0.314903\dots$ for both errors.

Decoupling introduced another source of information on the first spacing side. Bourgain and Watt [2] studied perturbed cones in connection with mean squares of the zeta function. The sharp cone square-function theorem of Guth, Wang, and Zhang [6] was formulated through wave envelopes: local averages of a square function at every intermediate angular scale. Guth and Maldague [5] refined that theorem by retaining an amplitude threshold. Their refinement implies small-cap decoupling for the cone. Li and Yang [11] combined this small-cap decoupling theorem with local arithmetic fourth moments and Huxley’s second spacing estimates, obtaining

$$R(X), \Delta(X) \ll_{\varepsilon} X^{\theta_* + \varepsilon}, \quad \theta_* = 0.3144831759741\dots \quad (1.3)$$

The improvement below is within this first-spacing architecture. It does not change the resonance-curve theorem or recover phases already removed from the short-block sum.

Theorem 1.1. *Put*

$$\vartheta = \frac{33333 - 8\sqrt{64515}}{99856} = 0.3134615421015514368\dots \quad (1.4)$$

For every $\varepsilon > 0$,

$$R(X) \ll_{\varepsilon} X^{\vartheta + \varepsilon}, \quad \Delta(X) \ll_{\varepsilon} X^{\vartheta + \varepsilon}. \quad (1.5)$$

The exact value in (1.4) comes from an interior maximum in the parameter $\log_T(H/M)$, not from the largest allowed value of H . We prove the inequality at that maximum by an algebraic square identity; the decimal approximations and the figures play no role in the proof.

1.2. The first-spacing estimate. Write $e(t) = \exp(2\pi it)$ and $k \sim K$ for integers $K \leq k < 2K$. For $1 \leq L \leq K$ and an array $a = (a_{kl})$ with $|a_{kl}| \leq 1$, define

$$F_a(x_1, x_2, x_3) = \sum_{k \sim K} \sum_{l \sim L} a_{kl} e(lx_1 + klx_2 + l\sqrt{k}x_3). \quad (1.6)$$

Let

$$\mathcal{G}_q(K, L)^q = \sup_{|a_{kl}| \leq 1} \frac{1}{2\sqrt{K}} \int_{-\sqrt{K}}^{\sqrt{K}} \int_0^1 \int_0^1 |F_a(x)|^q dx_1 dx_2 dx_3. \quad (1.7)$$

The supremum is convenient when this quantity is inserted into a large-sieve inequality. Every estimate for it is, in particular, uniform in each individual coefficient array.

Theorem 1.2. *Let $1 \leq L \leq K$ and $4 < q \leq 9/2$. For every $\varepsilon > 0$, with $P = KL$,*

$$\begin{aligned} \mathcal{G}_q(K, L)^q &\ll_{q, \varepsilon} P^{\varepsilon} [P^{q/2} + KL^{2q-5} + K^{q-2} L^{(q-1)/2} \\ &\quad + K^{3q/4-2} L^{5q/4-2}]. \end{aligned} \quad (1.8)$$

The constants are uniform when q belongs to a compact subinterval of $(4, 9/2]$.

Corollary 1.3. *Under the hypotheses of Theorem 1.2, if*

$$K^{q-4} \leq L, \quad L^{3q-8} \leq K^{8-q}, \quad (1.9)$$

then

$$\mathcal{G}_q(K, L) \ll_{q, \varepsilon} (KL)^{1/2+\varepsilon}. \quad (1.10)$$

In particular,

$$L^{11/7} \leq K \leq L^2 \implies \mathcal{G}_{9/2}(K, L) \ll_{\varepsilon} (KL)^{1/2+\varepsilon}. \quad (1.11)$$

Proof. Divide the third term of (1.8) by $P^{q/2}$; the quotient is $(K^{q-4}/L)^{1/2}$. For the fourth term the quotient is $(L^{3q-8}/K^{8-q})^{1/4}$. For the second term it is

$$K^{-(q-2)/2} L^{(3q-10)/2}.$$

The second condition in (1.9) makes this at most one, since

$$(3q-8)(q-2) - (3q-10)(8-q) = 6(q-4)^2 \geq 0.$$

The remaining assertion follows by setting $q = 9/2$. $\square$

For unit-modulus arrays, orthogonality in x_1, x_2 gives $\|F_a\|_{L_2^\#}^2 \asymp KL$. Thus a bound uniform in such arrays cannot improve the square-root order in (1.10). The point is to enlarge the range in which that order is available.

1.3. Why the amplitude decomposition matters. Under the change of physical variables

$$z_1 = Lx_1, \quad z_2 = L\sqrt{K}x_3, \quad z_3 = KLx_2,$$

the frequencies become

$$\xi_{kl} = \left(\frac{l}{L}, \frac{l\sqrt{k}}{L\sqrt{K}}, \frac{lk}{LK} \right) = r(1, t, t^2), \quad r = \frac{l}{L}, \quad t = \sqrt{k/K}. \quad (1.12)$$

The integer k therefore labels a generator of a cone, while l varies along that generator. We call terms with the same k *same-ray terms*; terms pairing distinct k 's are *cross-ray terms*.

Li and Yang use small-cap decoupling and, on each small cap, an inequality of the form

$$\|F_\sigma\|_q^q \leq \|F_\sigma\|_\infty^{q-4} \|F_\sigma\|_4^4. \quad (1.13)$$

The fourth moment and the largest pointwise amplitude in this inequality need not arise from the same configurations. We instead use [5, Theorem 2] before passing to a small-cap decoupling inequality. At angular scale s , this theorem restricts a contributing wave envelope U at amplitude α by a lower bound of the form

$$M_U \gtrsim_\varepsilon P^{-\varepsilon} \alpha^2 s^2,$$

where M_U is a normalized local square-function mass.

Inside this mass we make the exact decomposition

$$M_U = D_U + C_U.$$

Here $D_U \geq 0$ contains the same-ray squares and $C_U \in \mathbb{R}$ is the cross-ray contribution. Both terms have fourth-moment bounds, but their pointwise upper bounds differ:

$$D_U \lesssim sP \left(1 + \frac{1}{Ks^2} \right), \quad (1.14)$$

$$M_U, |C_U| \lesssim sP \left(1 + \frac{1}{Ks^2} \right) \left(1 + \frac{1}{Ls} \right). \quad (1.15)$$

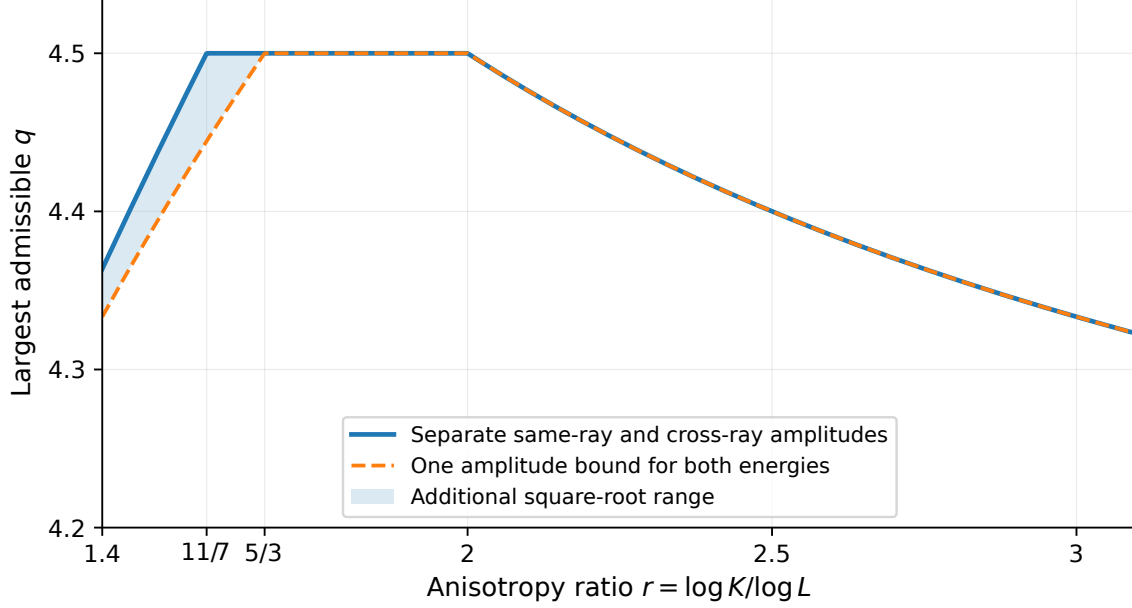


FIGURE 1. The square-root region, expressed using $r = \log K / \log L$ for $L > 1$. The upper boundary from Corollary 1.3 is $\min\{9/2, 4 + 1/r, 8(r+1)/(r+3)\}$. The dashed boundary is what the same envelope calculation gives if the radial energy is assigned the total amplitude bound. Both comparisons use the same arithmetic fourth-moment information; the difference is the amplitude assigned to the same-ray contribution.

The large radial degeneracy KL^3 occurs only in the energy of D_U . Pairing it with the larger upper bound for M_U unnecessarily charges it for coherence between distinct rays. Keeping the two amplitude thresholds separate replaces the contribution

$$K^{q-3}L^{q-1} \quad \text{by} \quad K^{3q/4-2}L^{5q/4-2}.$$

The ratio is $(L/K)^{(q-4)/4}$. The full proof keeps the sign of C_U until a pointwise splitting inequality is applied; no positivity of C_U is assumed.

1.4. Organization and external inputs. Section 2 implements the discrete frequency localization with the exact angular projections used by Guth and Maldague. Two rotated angular grids suffice, and an entire k -ray is assigned to one grid. This avoids cutting a Fourier atom with an angular multiplier. Section 3 proves the local density and arithmetic energy estimates, including the cross-ray count with the all-equal- k configurations removed. Section 4 integrates the two amplitude bounds and proves Theorem 1.2.

Sections 5 and 6 insert the four-term estimate into the Li–Yang large-sieve interface, retain all denominator scales, and check both admissible block lengths. Section 7 proves the Fourier and hyperbola reductions, with the circle reduction derived through the character modulo four. The analytic input is [5, Theorem 2]; the arithmetic input is [11, Lemma 4.4], with its stated block construction. The classical one-dimensional estimates and the finite sawtooth approximation are recorded explicitly where used. No higher-degree Weyl inverse theorem is an input to the proof.

2. DISCRETE LOCALIZATION FOR THE WAVE-ENVELOPE THEOREM

2.1. Notation and a common cutoff. All Fourier transforms use the convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^3} f(z) e(-z \cdot \xi) dz.$$

A factor P^ε may absorb a fixed number of logarithmic factors. In arguments using the wave-envelope theorem we first choose its small exponent $\delta > 0$ and then take δ sufficiently small in terms of the final ε . All constants in geometric comparisons below are independent of K, L, P, s and of the centers of the envelopes.

We can assume that P exceeds a fixed absolute constant. Define the unlocalized normalized sum

$$F_0(z) = \sum_{k \sim K, l \sim L} a_{kl} e(\xi_{kl} \cdot z),$$

with ξ_{kl} as in (1.12). The box defining (1.7) becomes

$$[0, L] \times [-P, P] \times [0, P].$$

The function F_0 is periodic in z_1 with period L . Repeating this period $\lceil K \rceil$ times produces a box of volume comparable to P^3 with exactly the same normalized integral. After division by P the enlarged box lies in a fixed compact set.

Choose once and for all a Schwartz function χ such that $\widehat{\chi}$ is supported in a sufficiently small ball $B(0, c)$ and $|\chi|$ is bounded below on this fixed compact set. Such a function is obtained by taking the inverse Fourier transform of a nonnegative even smooth bump supported in a sufficiently small ball. The small support makes that inverse transform positive on the prescribed compact set. In particular, no compactly supported Fourier transform is asserted to have an inverse transform equal to one on an open set.

Set

$$f(z) = \chi(z/P) F_0(z). \quad (2.1)$$

It follows that, for the array under consideration,

$$\|F_a\|_{L_{\#}^q}^q \lesssim P^{-3} \int_{\mathbb{R}^3} |f(z)|^q dz. \quad (2.2)$$

Every Fourier atom of f is supported in a ball of radius c/P about one of the points ξ_{kl} .

2.2. The exact cone and angular projections. Let

$$A(u, v, w) = \frac{3}{5} \left(v, \frac{w-u}{2}, \frac{w+u}{2} \right).$$

This fixed invertible map sends the frequencies to the circular cone. Indeed,

$$(A\xi_{kl})_1^2 + (A\xi_{kl})_2^2 = (A\xi_{kl})_3^2, \quad \frac{3}{5} \leq (A\xi_{kl})_3 < \frac{9}{5}.$$

The image is strictly inside the fixed radial truncation $1/2 < \xi_3 < 2$ used in [5, Section 5]. Choose $\mathcal{R} \in 4^{\mathbb{N}}$ so that $P \leq \mathcal{R} < 4P$. By choosing c sufficiently small, all transformed Fourier atoms lie in the $\mathcal{R}^{-1}$ neighborhood of that cone.

In the notation of [5, Section 5], canonical angular caps are defined in cylindrical coordinates by a partition of the angle into intervals of length

$$\delta_0 = 2\pi \mathcal{R}^{-1/2}.$$

The projections in the theorem are sharp Fourier restrictions to these sectors. This exact definition permits the following localization.

Lemma 2.1. *The function f in (2.1) is the sum of two functions such that, for each one, after a rotation of the circular cone, every canonical projection has the form*

$$f_\theta(z) = \chi(z/P) \sum_{k \in \mathcal{K}_\theta} \sum_{l \sim L} a_{kl} e(\xi_{kl} \cdot z). \quad (2.3)$$

The sets $\mathcal{K}_\theta$ are disjoint. An entire ray, namely all the frequencies with a given k , is assigned to the same angular grid and the same canonical cap. No parameter-dependent multiplier norm is required.

Proof. Consider the angular grid with endpoints $j\delta_0$ and the grid with endpoints $(j + 1/2)\delta_0$. Every angle is at distance at least $\delta_0/4$ from the endpoints of one of these two grids. The circular angle of $A\xi_{kl}$ depends only on k , not on l . Assign the whole k -ray to a grid having this separation property.

The angular diameter of a transformed Fourier atom is $O(c/P)$, uniformly in k, l , because its radial coordinate is bounded away from zero. This is smaller than $\delta_0/8$ for all sufficiently large P . Thus each atom lies wholly in the interior of a single canonical angular sector of its assigned grid. A sharp projection to that sector either keeps the entire atom or removes it. This proves (2.3).

The second grid is the first grid rotated by $\delta_0/2$. The circular cone, its radial truncation, and the constants in the wave-envelope theorem are invariant under this rotation. Its dependence on P is harmless because the rotation is orthogonal. The fixed map A changes integrals and envelope volumes by fixed determinant factors, which cancel in normalized local averages. Finally,

$$\int |f^{(0)} + f^{(1)}|^q \leq 2^{q-1} \left(\int |f^{(0)}|^q + \int |f^{(1)}|^q \right).$$

It suffices to estimate either class with the same uniform bound. The finitely many bounded values of P are included by increasing the constant. $\square$

We henceforth work with one of these two classes and suppress the superscript. Coarse angular sectors are unions of canonical sectors in the same grid, exactly as in [5, Section 5]. Since all atom supports avoid the fine boundaries, they also avoid every coarse boundary. The corresponding projections therefore keep whole atoms as well.

2.3. Envelopes, weights, and the external theorem. At a dyadic angular scale $\mathcal{R}^{-1/2} \leq s \leq 1$, let τ be a coarse sector. Its wave envelopes have dimensions comparable to

$$Ps^2 \times Ps \times P. \quad (2.4)$$

In circular coordinates these dimensions are measured in the generator, angular tangent, and normal directions. We pull the envelopes and their weights back to the z coordinates. They may then be parallelepipeds rather than orthogonal boxes; all volume and Fourier comparisons below are unchanged by this fixed linear map.

For each τ , the envelopes tile $\mathbb{R}^3$ by translation. We use the weights W_U from [5, Sections 2 and 5]. In normalized envelope coordinates they are fixed constant multiples of

$$W(u) = \prod_{j=1}^3 (1 + u_j^2)^{-100}. \quad (2.5)$$

They are nonnegative, satisfy $\int W_U \asymp |U|$, and have bounded overlap:

$$\sum_{U \parallel U_{\tau, P}} W_U(z) \lesssim 1. \quad (2.6)$$

All derivatives of W are integrable. The same assertions hold after the fixed coordinate changes already made.

Define

$$M_U = \frac{1}{|U|} \int_{\mathbb{R}^3} \sum_{\theta \subset \tau} |f_\theta(z)|^2 W_U(z) dz. \quad (2.7)$$

The amplitude-dependent wave-envelope theorem [5, Theorem 2] gives, for every $\delta > 0$ and $\alpha > 0$,

$$\alpha^4 |\{z : |f(z)| > \alpha\}| \leq C_\delta P^\delta \sum_s \sum_\tau \sum_{U \in \mathcal{U}_\tau(\alpha)} |U| M_U^2, \quad (2.8)$$

where a contributing envelope satisfies

$$C_\delta P^\delta M_U \geq \frac{\alpha^2}{n_s^2}, \quad n_s = \#\{\tau : f_\tau \not\equiv 0\} \lesssim s^{-1}. \quad (2.9)$$

Here n_s counts the active coarse caps at that scale; it is not the number of fine caps inside a given coarse cap. Adding the two endpoint scales to the sum only enlarges the right-hand side. We will use (2.8) in precisely this form.

2.4. The adapted Fourier metric. Fix a sector τ and choose a parameter t_0 at its angular endpoint. On its support, $t_0 \asymp 1$ and $|t - t_0| \lesssim s$. The following linear coordinates on frequency space are useful:

$$\eta_0 = r, \quad \eta_1 = r(t - t_0), \quad \eta_2 = r(t - t_0)^2. \quad (2.10)$$

They are linear functions of (r, rt, rt^2) . If B_U maps a fixed unit cube to the centered envelope, then its Fourier-dual metric is comparable to

$$|B_U^\top \nu| \asymp P s^2 |\eta_0(\nu)| + P s |\eta_1(\nu)| + P |\eta_2(\nu)|. \quad (2.11)$$

Here η_j is applied as a linear functional to a difference vector ν .

For completeness, omit the harmless scalar $3/5$ in A , put $D_0 = 1 + t_0^2$, and write the image of $r(1, t, t^2)$ in the orthogonal circular frame consisting of the generator, angular tangent, and normal at t_0 . The coefficients in that frame are

$$\begin{aligned} a_c &= \frac{D_0}{2} \eta_0 + t_0 \eta_1 + \frac{t_0^2}{2D_0} \eta_2, \\ a_t &= \eta_1 + \frac{t_0}{D_0} \eta_2, \quad a_n = -\frac{1}{2D_0} \eta_2. \end{aligned} \quad (2.12)$$

This is a triangular invertible change of coordinates with uniformly bounded coefficients and inverse coefficients. Weighting its rows by $P s^2, P s, P$, respectively, preserves boundedness of both maps because $s \leq 1$. This proves (2.11).

Lemma 2.2. *For every fixed $A_0 > 0$, uniformly in U ,*

$$\left| |\chi(\cdot/\widehat{P})|^2 W_U(\nu) \right| \leq C_{A_0} |U| (1 + |B_U^\top \nu|)^{-A_0}. \quad (2.13)$$

Proof. Write $z = z_U + B_U u$ in the Fourier integral. The operator norm of B_U/P is bounded, since the largest envelope dimension is $O(P)$. Every derivative in u of $|\chi((z_U + B_U u)/P)|^2$ is consequently bounded uniformly in z_U, U, P . After multiplication by the fixed weight (2.5), the derivatives are integrable, with uniform norms. Repeated integration by parts in u , followed by the Jacobian factor $|\det B_U| \asymp |U|$, gives (2.13). $\square$

This estimate is for the actual weights in (2.8). Exact integer orthogonality will be used only after those weights have been summed out. We do not claim that a single envelope integral retains that orthogonality.

3. SAME-RAY AND CROSS-RAY ENERGIES

3.1. Local frequency density. For each retained k , put

$$f_k(z) = \chi(z/P) \sum_{l \sim L} a_{kl} e(\xi_{kl} \cdot z). \quad (3.1)$$

By Lemma 2.1, $f_\theta = \sum_{k \in \mathcal{K}_\theta} f_k$. Define

$$D_U = \frac{1}{|U|} \int \sum_{k \in \tau} |f_k|^2 W_U, \quad (3.2)$$

$$C_U = M_U - D_U. \quad (3.3)$$

Both quantities are real, and $D_U \geq 0$; C_U is not assumed nonnegative. Set

$$\mathcal{M}(s) = sP \left(1 + \frac{1}{Ls}\right) \left(1 + \frac{1}{Ks^2}\right), \quad (3.4)$$

$$\mathcal{D}(s) = sP \left(1 + \frac{1}{Ks^2}\right). \quad (3.5)$$

Lemma 3.1. *At every admissible scale s and every envelope U ,*

$$M_U \lesssim \mathcal{M}(s), \quad D_U \lesssim \mathcal{D}(s), \quad |C_U| \lesssim \mathcal{M}(s). \quad (3.6)$$

Also, at each fixed scale,

$$P^{-3} \sum_{\tau, U} |U| M_U \lesssim P, \quad P^{-3} \sum_{\tau, U} |U| D_U \lesssim P. \quad (3.7)$$

Proof. For two frequencies in τ whose difference lies in the 2^j dilation of the dual envelope, (2.11) implies

$$|r - r'| \lesssim \frac{2^j}{Ps^2}, \quad |r(t - t_0) - r'(t' - t_0)| \lesssim \frac{2^j}{Ps}.$$

Since $r, r' \asymp 1$ and $|t - t_0|, |t' - t_0| \lesssim s$, the second inequality and the first give $|t - t'| \lesssim 2^j/(Ps)$. Thus

$$|l - l'| \lesssim \frac{2^j}{Ks^2}, \quad |k - k'| \lesssim \frac{2^j}{Ls}. \quad (3.8)$$

For a fixed frequency there are therefore at most

$$C \left(1 + \frac{2^j}{Ks^2}\right) \left(1 + \frac{2^j}{Ls}\right)$$

neighbors at that dilation. The number of frequencies in τ is $O(sP)$: its k interval has length $O(sK)$, and $sK \gtrsim \sqrt{K/L} \gtrsim 1$ up to absolute constants.

Expand (2.7), use Lemma 2.2, and sum these neighbor counts against $2^{-A_0 j}$ with $A_0 > 4$. Taking the absolute value of each pair only increases the upper bound. This proves $M_U \lesssim \mathcal{M}(s)$. For D_U , all pairs have $k = k'$, so the second factor in the neighbor count is absent. This proves $D_U \lesssim \mathcal{D}(s)$. Finally, $|C_U| \leq M_U + D_U \lesssim \mathcal{M}(s)$.

To prove (3.7), first note the exact orthogonality

$$P^{-3} \int \sum_{\theta} |f_\theta|^2 \lesssim P, \quad P^{-3} \int \sum_k |f_k|^2 \lesssim P. \quad (3.9)$$

Indeed, distinct integer pairs have distinct first or third normalized coordinates, separated by at least $1/P$. The Fourier transform of $|\chi(\cdot/P)|^2$ is supported in $B(0, 2c/P)$, so all off-diagonal integrals vanish when $2c < 1$. The diagonal integrals equal a fixed constant times $P^3 \sum |a_{kl}|^2 \lesssim P^4$. Each fine cap or ray belongs to one coarse sector at a fixed scale. Sum the envelope weights using (2.6) and apply (3.9). $\square$

Define the two energy sums

$$\mathcal{E}_D(s) = P^{-3} \sum_{\tau, U} |U| D_U^2, \quad \mathcal{E}_C(s) = P^{-3} \sum_{\tau, U} |U| |C_U|^2. \quad (3.10)$$

The density estimates immediately imply

$$\mathcal{E}_D(s) \lesssim P\mathcal{D}(s), \quad \mathcal{E}_C(s) \lesssim P\mathcal{M}(s). \quad (3.11)$$

For the second assertion use $|C_U|^2 \leq 2M_U^2 + 2D_U^2$, followed by Lemma 3.1. The nonnegativity of C_U is not needed.

3.2. A fourth-moment count without the radial degeneracy.

Lemma 3.2. *Let $1 \leq V \leq CK$. The number of integer quadruples $(k_i, l_i)_{i=1}^4$ with $k_i \sim K$, $l_i \sim L$,*

$$l_1 + l_2 = l_3 + l_4, \quad (3.12)$$

$$k_1 l_1 + k_2 l_2 = k_3 l_3 + k_4 l_4, \quad (3.13)$$

with $\text{diam}\{k_1, k_2, k_3, k_4\} \leq V$ and with the k_i not all equal, is

$$\ll_{\varepsilon} P^{\varepsilon} K V^2 L^2. \quad (3.14)$$

If the restriction that the k_i are not all equal is removed, an additional $O(KL^3)$ suffices.

Proof. Fix k_4 and write

$$d_i = k_i - k_4 \quad (1 \leq i \leq 3), \quad a = l_1 - l_3, \quad b = l_2 - l_3.$$

Then $|d_i| \lesssim V$, $|a|, |b| \lesssim L$, and $l_4 = l_3 + a + b$. The second equation becomes

$$ad_1 + bd_2 = l_3(d_3 - d_1 - d_2). \quad (3.15)$$

If $ad_1 + bd_2 \neq 0$, choose k_4, d_1, d_2, a, b in $O(KV^2L^2)$ ways. The left side is a nonzero integer of magnitude $O(LV)$. Its positive divisor l_3 has $O_{\varepsilon}(P^{\varepsilon})$ choices, and then d_3 is determined. Dropping range restrictions only enlarges the count.

If $ad_1 + bd_2 = 0$, positivity of l_3 forces $d_3 = d_1 + d_2$. If $ad_1 = -bd_2 \neq 0$, choose a, d_1 and then use the divisor bound to choose b, d_2 ; choose l_3, k_4 freely. This contributes $O_{\varepsilon}(P^{\varepsilon} K V L^2)$.

The remaining zero-product alternatives are covered by

$$a = b = 0, \quad a = d_2 = 0, \quad d_1 = b = 0, \quad d_1 = d_2 = 0.$$

Their first three contributions are at most $O(KLV^2)$, $O(KVL^2)$, and $O(KVL^2)$, respectively. In the last alternative, (3.15) forces $d_3 = 0$, so every $k_i = k_4$. This alternative is excluded. As $V, L \geq 1$, all retained contributions are bounded by (3.14). When the exclusion is dropped, the all-equal- k family has $O(KL^3)$ elements by the first equation in (3.12). $\square$

3.3. The two arithmetic energies.

Lemma 3.3. *At every scale,*

$$\mathcal{E}_D(s) \lesssim sP^2 + KL^3, \quad (3.16)$$

$$\mathcal{E}_C(s) \ll_{\varepsilon} P^{\varepsilon} P s^2 K^2 L. \quad (3.17)$$

Consequently,

$$\mathcal{E}_D(s) \ll_{\varepsilon} P^{\varepsilon} \min\{P\mathcal{D}(s), sP^2 + KL^3\}, \quad (3.18)$$

$$\mathcal{E}_C(s) \ll_{\varepsilon} P^{\varepsilon} \min\{P\mathcal{M}(s), P s^2 K^2 L\}. \quad (3.19)$$

Proof. Put

$$d_\tau(z) = \sum_{k \in \tau} |f_k(z)|^2, \quad c_\tau(z) = \sum_{\theta \subset \tau} |f_\theta(z)|^2 - d_\tau(z).$$

Weighted Cauchy–Schwarz and (2.6) give

$$\mathcal{E}_D(s) \lesssim P^{-3} \sum_{\tau} \int |d_\tau(z)|^2 dz, \quad (3.20)$$

$$\mathcal{E}_C(s) \lesssim P^{-3} \sum_{\tau} \int |c_\tau(z)|^2 dz. \quad (3.21)$$

This is where the weights are removed. The common factor in each fourth-moment integrand is $|\chi(z/P)|^4$, whose Fourier transform is supported in $B(0, 4c/P)$. A nonzero integral of a frequency quadruple thus requires

$$\left| \frac{l_1 + l_2 - l_3 - l_4}{L} \right| < \frac{4c}{P}, \quad \left| \frac{k_1 l_1 + k_2 l_2 - k_3 l_3 - k_4 l_4}{P} \right| < \frac{4c}{P}.$$

For $4c < 1$ the numerators, which are integers, must vanish. The remaining approximate square-root equation may be discarded in an upper bound, and every normalized integral has absolute value $O(1)$.

For d_τ^2 the quadruples consist of one difference on a ray k and another on a ray k' . If $k \neq k'$, the two integer equations force the two l -differences to vanish separately. There are $O(sK^2)$ choices for k, k' in a common coarse sector and $O(L^2)$ choices for the radial diagonal. If $k = k'$, the radial additive relation has $O(L^3)$ solutions. Summing over sectors gives $O(sK^2L^2 + KL^3)$, proving (3.16).

Every term of c_τ pairs distinct rays lying in the same canonical cap. Hence, in the expansion of $|c_\tau|^2$, the four k 's cannot all be equal. All four belong to the same coarse sector, so their diameter is $O(sK)$. Each quadruple is counted in at most a fixed number of coarse sectors. Apply Lemma 3.2 with $V \asymp sK$ (using $1 + sK \asymp sK$ if necessary). The result is

$$P^\varepsilon K(sK)^2 L^2 = P^\varepsilon P s^2 K^2 L,$$

which proves (3.17). Taking absolute values after the expansion is legitimate; no sign of the individual coefficient products is asserted. Combine these estimates with (3.11) to obtain (3.18) and (3.19). $\square$

4. AMPLITUDE SPLITTING AND THE FIRST-SPACING BOUND

4.1. A restricted splitting inequality.

Lemma 4.1. *If $M \geq 0$, $D \geq 0$, $C \in \mathbb{R}$, and $M = D + C$, then for every $u > 0$,*

$$M^2 \mathbf{1}_{\{M \geq u\}} \leq 4D^2 \mathbf{1}_{\{D \geq u/2\}} + 4|C|^2 \mathbf{1}_{\{|C| \geq u/2\}}. \quad (4.1)$$

Proof. If $M < u$ there is nothing to prove. Otherwise $\max(D, |C|) \geq M/2 \geq u/2$, because $M \leq D + |C|$. The corresponding summand on the right is at least M^2 . $\square$

Proposition 4.2. *Let $4 < q \leq 9/2$ and put $\rho = (q - 4)/2$. Then*

$$\begin{aligned} \mathcal{G}_q(K, L)^q &\ll_{q, \varepsilon} P^\varepsilon \sum_s \left[\min\{P\mathcal{D}(s), sP^2 + KL^3\} \left(\frac{\mathcal{D}(s)}{s^2} \right)^\rho \right. \\ &\quad \left. + \min\{P\mathcal{M}(s), Ps^2K^2L\} \left(\frac{\mathcal{M}(s)}{s^2} \right)^\rho \right]. \end{aligned} \quad (4.2)$$

Proof. From (2.9), a contributing envelope obeys $M_U \geq c_\delta P^{-\delta} \alpha^2 s^2$. Apply Lemma 4.1 to $M_U = D_U + C_U$ at this threshold. The same-ray contribution can occur only if

$$D_U \geq \frac{1}{2} c_\delta P^{-\delta} \alpha^2 s^2, \quad \text{hence} \quad \alpha \lesssim_\delta P^{\delta/2} \frac{\mathcal{D}(s)^{1/2}}{s}.$$

The cross-ray contribution similarly requires

$$\alpha \lesssim_\delta P^{\delta/2} \frac{\mathcal{M}(s)^{1/2}}{s}.$$

Use the layer-cake identity

$$\int |f|^q = q \int_0^\infty \alpha^{q-1} |\{|f| > \alpha\}| d\alpha$$

and insert (2.8) with the split right-hand side. Tonelli's theorem applies to the resulting nonnegative majorants. At a fixed scale, the remaining amplitude integral is $\int_0^{A_s} \alpha^{q-5} d\alpha = A_s^{q-4}/(q-4)$, with the appropriate bound for A_s in each branch. Divide by P^3 , use Lemma 3.3, and then use (2.2). Choose δ small enough to absorb the factors P^δ and $P^{\delta(q-4)/2}$ into the displayed P^ε . Finally sum the two angular classes from Lemma 2.1. This proves (4.2). $\square$

4.2. The same-ray contribution. We estimate its summand first. If $s \geq K^{-1/2}$, then $\mathcal{D}(s) \lesssim sP$. The density energy gives

$$P\mathcal{D}(s) \left(\frac{\mathcal{D}(s)}{s^2} \right)^\rho \lesssim P^{2+\rho} s^{1-\rho} \leq P^{q/2}. \quad (4.3)$$

If $P^{-1/2} \lesssim s \leq K^{-1/2}$, then $\mathcal{D}(s) \lesssim L/s$. Use the arithmetic energy instead. Its sP^2 term is

$$P^2 L^\rho s^{1-3\rho}.$$

Since $0 < \rho \leq 1/4$, this is largest at $s = K^{-1/2}$. Dividing that endpoint value by $P^{2+\rho}$ gives $K^{(\rho-1)/2} \leq 1$. The KL^3 term is

$$KL^3 L^\rho s^{-3\rho} \lesssim KL^{3+\rho} P^{3\rho/2} = K^{3q/4-2} L^{5q/4-2}. \quad (4.4)$$

Thus the same-ray part is controlled, at every scale, by the first and fourth terms of (1.8).

4.3. The cross-ray contribution. There are three angular regimes. If

$$s \geq \max(L^{-1}, K^{-1/2}),$$

then $\mathcal{M}(s) \lesssim sP$, and the density energy gives

$$P\mathcal{M}(s) \left(\frac{\mathcal{M}(s)}{s^2} \right)^\rho \lesssim P^{q/2} s^{3-q/2} \leq P^{q/2}. \quad (4.5)$$

When $K \leq L^2$, the intermediate range is $L^{-1} \leq s \leq K^{-1/2}$. In that range $\mathcal{M}(s) \lesssim L/s$, so the same density argument gives

$$PL^{(q-2)/2} s^{5-3q/2} \leq PL^{2q-6} = KL^{2q-5}.$$

The power of s is negative and the largest value is at $s = L^{-1}$. When $K \geq L^2$, the intermediate range is $K^{-1/2} \leq s \leq L^{-1}$. Then $\mathcal{M}(s) \lesssim K$, and the contribution is at most

$$PK^{(q-2)/2} s^{-(q-4)} \leq PK^{q-3} \leq K^{q-2} L^{(q-1)/2}.$$

Finally suppose

$$P^{-1/2} \lesssim s \leq \min(L^{-1}, K^{-1/2}).$$

Here $\mathcal{M}(s) \lesssim s^{-2}$, and the summand is at most

$$P \min\{s^{6-2q}, K^2 L s^{10-2q}\}. \quad (4.6)$$

The first function in the minimum is decreasing in s and the second is increasing. They agree at

$$s_c = K^{-1/2} L^{-1/4}. \quad (4.7)$$

Whether or not this scale lies in the interval under consideration, the maximum over all positive s is an upper bound. Its value is

$$Ps_c^{6-2q} = K^{q-2} L^{(q-1)/2}.$$

This gives the third term of (1.8).

Proof of Theorem 1.2. Combine (4.3)–(4.6) in Proposition 4.2. There are $O(\log(2P))$ scales, and their number is absorbed into P^ε . Adjust the auxiliary small exponents to obtain the stated value of ε . The construction and all estimates were uniform in $|a_{kl}| \leq 1$, so the supremum in (1.7) is permitted. $\square$

Remark 4.3. If the same-ray energy KL^3 were multiplied by the total cutoff $(\mathcal{M}(s)/s^2)^\rho$, its smallest-scale contribution would be $K^{q-3}L^{q-1}$. Equation (4.4) is smaller by $(L/K)^{(q-4)/4}$. No radial additive relation has been removed: the gain comes solely from assigning that relation its own amplitude bound. Figure 1 displays the resulting change in the square-root region.

5. THE ARITHMETIC INTERFACE AND BOTH BLOCK LENGTHS

5.1. The double sums and the quoted large sieve. For a real phase F on $[1, 2]$, write

$$S(H, M, T; F) = \sum_{h \sim H} g(h/H) \sum_{m \sim M} G(m/M) e\left(\frac{hT}{M} F(m/M)\right). \quad (5.1)$$

The bounded-variation norms of g, G are measured by $\|w\|_{\text{BV}} = \|w\|_\infty + \text{Var}(w)$. Constants below may depend on these norms and on fixed phase bounds. In particular, interval indicator weights give estimates uniform in both upper summation endpoints.

The arithmetic input is the following specialization of [11, Lemma 4.4, equations (4.8) and (4.12)]. We state sufficient hypotheses in the range $M \leq T^{1/2}$, which is all that will be used. Assume

$$c \leq |F^{(j)}(t)| \leq C \quad (j = 1, 2, 3), \quad |F'(t)F'''(t) - 3F''(t)^2| \geq c \quad (5.2)$$

throughout $[1, 2]$. There are fixed constants in the block construction, depending only on these bounds, with the following properties.

In Case A it suffices that

$$H \geq M^{-9}T^4(\log T)^{171/140}, \quad H \leq MT^{-49/164}, \quad (5.3)$$

use a block length comparable to

$$N_A = M^{41/25}H^{-16/25}T^{-49/100}(\log T)^{969/14000}. \quad (5.4)$$

In Case B it suffices that

$$H \leq \min\{M^{35/69}T^{-2/23}, B_0M^{3/2}T^{-1/2}\}, \quad M \leq T^{1/2}, \quad (5.5)$$

where $B_0 > 0$ is a fixed phase-dependent constant; use a block length comparable to

$$N_B = \min\left\{M^{7/8}H^{-29/40}T^{-3/20}(\log T)^{969/5600}, \quad M^2H^{-1/3}T^{-2/3}\right\}. \quad (5.6)$$

The fixed small factors and integer rounding in these lengths are those allowed by the cited construction. Comparisons of such lengths below are understood up to those fixed factors.

For the chosen N , put

$$R_0^2 \asymp \frac{M^3}{NT}, \quad K \asymp \frac{NQ}{R_0^2}, \quad L \asymp \frac{HQ}{R_0^2}, \quad \eta \asymp \frac{R_0^2}{NH}. \quad (5.7)$$

Let $G_q(K, L, \eta)$ denote the supremum of the normalized L^q norm of (1.6) over complete periods in x_1, x_2 and

$$|x_3| \leq (\eta L \sqrt{K})^{-1}.$$

Then the large-sieve input is

$$|S| \ll_\varepsilon T^\varepsilon \max_{R_0 \leq Q \leq Q_2} \left(\frac{R_0}{Q}\right)^{3-6/q} \frac{MR_0}{N} \left(\frac{H}{R_0}\right)^{22/(17q)} G_q(K, L, \eta), \quad (5.8)$$

where

$$Q_2 = R_0(H/R_0)^{39/119}(\log(2H/R_0))^{-3/4}. \quad (5.9)$$

We use $4 < q \leq 9/2$. The parameters in the construction satisfy $R_0 \lesssim H \ll N$ and $1 \leq L \leq K \leq \eta^{-1} \leq KL$, with fixed harmless adjustments at endpoints.

The two substantive assertions being quoted here are the admissible block construction and (5.8); we do not reprove Huxley's resonance-curve theory. They precede the insertion of the first-spacing estimate in [11]. In particular, condition (4.6) of that paper is a hypothesis needed for its particular first-spacing bound, not an additional hypothesis on (5.8). Section 4.3 there obtains it when substituting that bound into Lemma 4.4.

The precursor [3] was withdrawn because of its first-spacing assertions. We do not use those assertions or its numerical conclusion. The arithmetic interface quoted above is the one explicitly restated and used in [11]; the replacement first-spacing estimate is proved in Sections 2–4 of the present paper.

5.2. Every denominator scale. Put

$$\nu = Q/R_0, \quad K_0 = N/R_0, \quad L_0 = H/R_0.$$

Then $K \asymp \nu K_0$, $L \asymp \nu L_0$, and $\eta KL \asymp \nu^2$. Comparing the actual integrals, before normalizing their volumes, gives

$$G_q(K, L, \eta)^q \lesssim \nu^2 \mathcal{G}_q(K, L)^q. \quad (5.10)$$

If fixed comparison constants make an interval slightly larger than the defining one in (1.7), cover it by a bounded number of translates. Translation in x_3 only multiplies a_{kl} by a unit-modulus factor, so the supremum gives the same bound. This also accounts for the fixed constants implicit in (5.7). Formula (5.10) is a volume comparison, not a monotonicity statement about normalized norms.

Insert Theorem 1.2 and (5.10) into (5.8). After raising to the q th power, the four monomials have the following total powers of ν :

$$8 - 2q, \quad 4 - q, \quad \frac{11 - 3q}{2}, \quad 4 - q. \quad (5.11)$$

They are nonpositive throughout the stated range. Hence each term is largest at $Q \asymp R_0$. If the square-root conditions hold at (K_0, L_0) , the result is

$$|S| \ll_\varepsilon T^\varepsilon M \sqrt{\frac{H}{N}} \left(\frac{H}{R_0} \right)^{22/(17q)}. \quad (5.12)$$

No square-root condition has been imposed at the larger values of Q : the complete four-term estimate was kept until after (5.11) was verified.

5.3. Dependence on the block length. For fixed H, M, T one has $R_0 \asymp N^{-1/2}$, $K_0 \asymp N^{3/2}$, and $L_0 \asymp N^{1/2}$. The prefactor in (5.8), at $Q \asymp R_0$, has power $-3/2 + 11/(17q)$ in N . The complete four endpoint terms consequently have powers

$$-\frac{1}{2} + \frac{11}{17q}, \quad -\frac{1}{2} - \frac{6}{17q}, \quad \frac{1}{4} - \frac{177}{68q}, \quad \frac{1}{4} - \frac{57}{17q}. \quad (5.13)$$

All are negative for $4 < q \leq 9/2$. It follows that an estimate at an admissible $N_B \gtrsim N_A$ may be weakened to its value at N_A , term by term. Only after this comparison will we use the square-root conditions at N_A .

6. THE EXPONENT OPTIMIZATION AND THE HARD RANGE

6.1. Elementary estimates and reciprocal phases. For the final reductions it suffices to consider the uniformly regular reciprocal class

$$F(t) = \frac{c_0}{t + a_0} + b_0, \quad 1 \leq t \leq 2, \quad (6.1)$$

where $|c_0|$ is bounded above and bounded away from zero, $|t + a_0|$ is bounded above and bounded away from zero, and $b_0 \in \mathbb{R}$ is arbitrary. The constants may depend on these fixed bounds, but not on b_0 . All derivative conditions (5.2) hold; in particular,

$$F'F''' - 3(F'')^2 = -\frac{6c_0^2}{(t + a_0)^6}.$$

This class permits parameters a_0, c_0 to vary in the indicated compact sets. It contains the shifted reciprocals arising in the circle reduction.

Lemma 6.1. *For this class, $M \leq T^{1/2}$, and $H \geq 1$, uniformly over subintervals and bounded-variation weights,*

$$|S|/H \ll (HT)^{2/7} + 1, \quad (6.2)$$

$$|S|/H \ll (HT/M)^{1/2} + T^{1/4}. \quad (6.3)$$

Proof. The classical exponent pair $(2/7, 4/7)$ applied to the inner reciprocal sum gives

$$\left| \sum_{m \in I} e\left(\frac{hT}{M} F(m/M)\right) \right| \ll (hT/M^2)^{2/7} M^{4/7} + \frac{M^2}{hT} \ll (hT)^{2/7} + 1.$$

The pair follows from $(0, 1)$ by the successive operations B, A, A, B ; the derivative hypotheses for reciprocal functions and these operations are given in [4, Chapters 2–3]. This application is uniform in a_0, c_0 in the stated compact sets. The additive constant contributes only a phase depending on h and does not affect the one-dimensional estimate. Sum over $h \sim H$ to obtain (6.2).

The second-derivative bound [4, Theorem 2.2], with $|\Phi_h''| \asymp hT/M^3$, gives

$$\left| \sum_{m \in I} e(\Phi_h(m)) \right| \ll (hT/M)^{1/2} + \frac{M^{3/2}}{(hT)^{1/2}} + 1.$$

The second term is at most $T^{1/4}$ when $M \leq T^{1/2}$ and $h \geq 1$. Summation in h proves (6.3). Partial summation supplies the stated weights. In particular, the reciprocal term in the second-derivative estimate has not been discarded without a bound. $\square$

We seek $|S|/H \ll_\varepsilon T^{\vartheta+\varepsilon}$ for $1 \leq H \leq MT^{-\vartheta}$ and $M \leq T^{1/2}$. By Lemma 6.1 only the range

$$T^{(7\vartheta-2)/2} < H \leq MT^{-\vartheta}, \quad H > MT^{2\vartheta-1}, \quad M \leq T^{1/2} \quad (6.4)$$

remains.

6.2. Checking the arithmetic cases. Let

$$B_* = M^{-9}T^4(\log T)^{171/140}.$$

If $H \geq B_*$, use Case A. Its upper condition holds because $\vartheta > 49/164$. Suppose instead that $H < B_*$. Interpolating $H \leq MT^{-\vartheta}$ and $H < B_*$ with weights $328/345$ and $17/345$ yields

$$H \leq M^{35/69}T^{(68-328\vartheta)/345}(\log T)^{969/16100}. \quad (6.5)$$

Since $\vartheta > 49/164$, the power of T is strictly smaller than $-2/23$, absorbing the logarithm. Also, by (6.4),

$$\frac{M^{3/2}T^{-1/2}}{H} \geq T^{(13\vartheta-4)/4} \longrightarrow \infty, \quad (6.6)$$

because $\vartheta > 4/13$. Thus the phase-dependent constant B_0 in (5.5) is allowed for sufficiently large T , and Case B applies.

To compare its length with N_A , use again (6.4):

$$H^{23}M^{27} \geq H^{50}T^{27\vartheta} \geq T^{202\vartheta-50} > T^{53/4}. \quad (6.7)$$

The last inequality has a fixed power margin, since $\vartheta > 253/808$. If $N_{B,1}$ and $N_{B,2}$ denote the two alternatives in (5.6), direct division gives

$$\frac{N_{B,1}}{N_A} = \left(\frac{B_*}{H}\right)^{17/200} \geq 1, \quad (6.8)$$

$$\frac{N_{B,2}}{N_A} = \left(\frac{H^{23}M^{27}}{T^{53/4}}\right)^{1/75} (\log T)^{-969/14000} \geq 1 \quad (6.9)$$

for all sufficiently large T . Fixed admissibility factors change these comparisons only by constants. By (5.13), the complete estimate at N_B is therefore bounded by its value at N_A .

6.3. The endpoint spacing parameters. Put $H/M = T^x$. In (6.4),

$$2\vartheta - 1 < x \leq -\vartheta. \quad (6.10)$$

At N_A , the powers of $K_0 = N_A/R_0$ and $L_0 = H/R_0$ are

$$\kappa(x) = -\frac{47}{200} - \frac{24x}{25}, \quad \lambda(x) = \frac{51}{200} + \frac{17x}{25}. \quad (6.11)$$

Logarithmic factors have been suppressed only in these power identities. They can either be retained in the four monomials, or absorbed by a smaller auxiliary ε .

Here $x > -3/8$, so $\lambda(x) > 0$ with a fixed margin; its smallest value is $(34/25)\vartheta - 17/40 > 0$. Moreover,

$$\log_T(N_A/H) = -\frac{41x}{25} - \frac{49}{100} + o(1) > 0, \quad \log_T(N_A/M) = -\frac{16x}{25} - \frac{49}{100} + o(1) < 0.$$

Thus $R_0 \ll H \ll N_A \ll M$, and $1 \ll L_0 \leq K_0$. Also $R_0 \gg 1$: from $M \geq HT^\vartheta$ and (6.4),

$$\log_T R_0 \geq \frac{9\vartheta - 2}{2} + \frac{8}{25}(2\vartheta - 1) - \frac{51}{200} + o(1) > 0.$$

These inequalities have fixed positive margins and supply the constant separations required in the block construction. At the actual Case B length, $N_B \gtrsim N_A$ implies $N_B/H \gg 1$ and $H/R_0 \gg 1$. The upper bound $N_B \leq N_{B,2}$ gives, separately,

$$\frac{N_B}{M} \lesssim MH^{-1/3}T^{-2/3} \leq T^{-1/6}H^{-1/3} \ll 1,$$

and

$$R_0^2 \gtrsim MH^{1/3}T^{-1/3} \geq T^{(17\vartheta-5)/3} \gg 1.$$

For the last inequality use $M \geq HT^\vartheta$ and $H > T^{(7\vartheta-2)/2}$. Thus neither the large block nor its associated spacing scale leaves the permitted range. Finally, $Q_2/H = (H/R_0)^{-80/119}(\log(2H/R_0))^{-3/4}$ tends to zero, so the displayed denominator range remains below the available H scale.

The major-arc term in the quoted construction has size $MR_0 \log H/\sqrt{HN}$. Compared to the right side of (5.12), its ratio, apart from a logarithm, is

$$\frac{R_0}{H}(H/R_0)^{-22/(17q)} \leq 1.$$

The other complementary terms are already included in the interface (5.8); no new gain is assigned to them.

6.4. A square identity for the largest exponent. The square-root conditions (1.9), at the level of powers of T , are

$$(q - 4)\kappa \leq \lambda, \quad (3q - 8)\lambda \leq (8 - q)\kappa. \quad (6.12)$$

Choose

$$q(x) = \begin{cases} 4 + \lambda(x)/\kappa(x), & x \leq -149/464, \\ 9/2, & x \geq -149/464. \end{cases} \quad (6.13)$$

Lemma 6.2. *For x in (6.10), the choice (6.13) lies in a fixed compact subinterval of $(4, 9/2]$ and satisfies (6.12).*

Proof. Let $r = \kappa/\lambda$. Both quantities are positive. The identity $r = 2$ occurs at $x = -149/464$. On the first interval $r \geq 2$ and $q = 4 + 1/r$. The first inequality is an equality; the second reduces to $4r^2 - 5r - 3 \geq 0$, valid for $r \geq 2$. On the second interval $q = 9/2$. Its two inequalities become $11/7 \leq r \leq 2$. The lower condition is equivalent to

$$x \leq -89/284,$$

and follows from $x \leq -\vartheta$ and $\vartheta > 89/284$. The lower endpoint $2\vartheta - 1$ is strictly larger than $-3/8$, at which λ vanishes. Compactness gives a fixed distance of $q(x)$ from 4. $\square$

Divide (5.12), evaluated at N_A , by H . Its exponent in T is

$$E(x, q) = \left(-\frac{9}{50} + \frac{22}{25q}\right)x + \frac{49}{200} + \frac{33}{100q}. \quad (6.14)$$

For the first choice of q this equals

$$E_1(x) = \frac{2208x^2 + 9396x + 1963}{40(632x + 137)}. \quad (6.15)$$

For the second it equals

$$E_2(x) = \frac{28x + 573}{1800}. \quad (6.16)$$

Define

$$x_0 = \frac{25280\vartheta - 9396}{4416} = -0.3332636357954664\dots \quad (6.17)$$

The following identity is exact:

$$\boxed{E_1(x) - \vartheta = \frac{276}{5} \frac{(x - x_0)^2}{632x + 137}.} \quad (6.18)$$

Indeed, after bringing $E_1(x) - \vartheta$ over its denominator, its numerator is

$$2208x^2 + (9396 - 25280\vartheta)x + 1963 - 5480\vartheta.$$

The value (1.4) makes its discriminant zero, and its double root is (6.17). This proves (6.18). The denominator is negative on (6.10); hence $E_1(x) \leq \vartheta$, with equality at $x = x_0$.

The function E_2 is increasing. Its largest value in the remaining interval is $E_2(-\vartheta)$, and

$$E_2(-\vartheta) \leq \vartheta \iff \vartheta \geq \frac{573}{1828}. \quad (6.19)$$

This inequality is strict. For example, direct rational squaring in (1.4) gives

$$\frac{313461}{10^6} < \vartheta < \frac{313462}{10^6}, \quad \frac{313457}{10^6} < \frac{573}{1828} < \frac{313458}{10^6}.$$

The same bounds verify all strict numerical inequalities on ϑ used above. They also give $2\vartheta - 1 < x_0 < -149/464 < -\vartheta$.

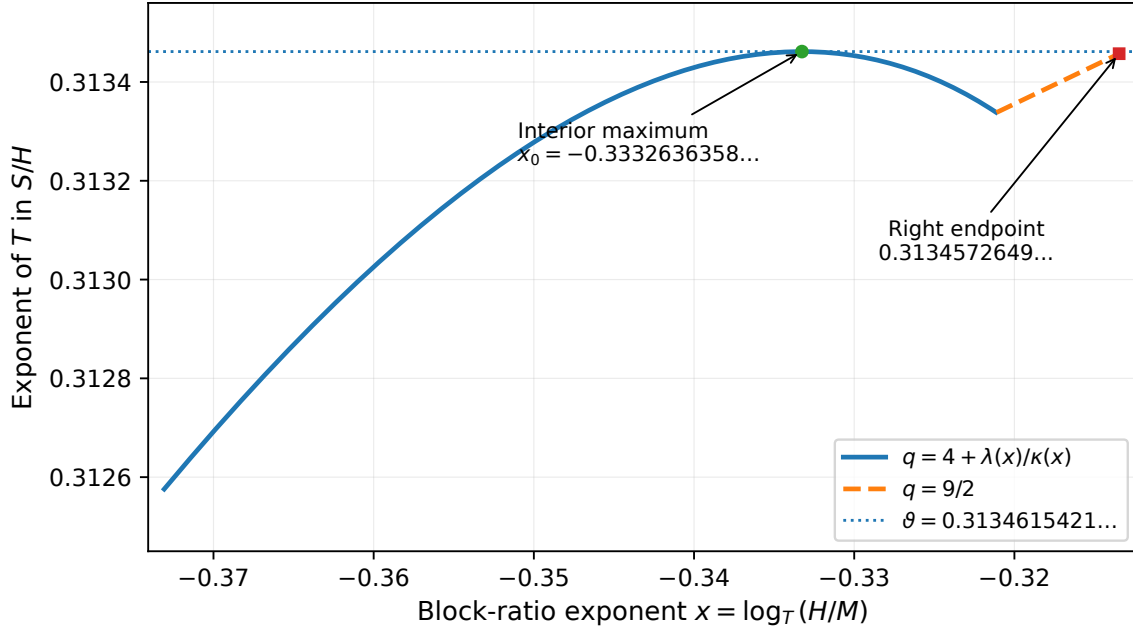


FIGURE 2. The exponent obtained from (6.13). The first piece reaches its maximum at the interior point x_0 ; the increasing $q = 9/2$ piece has a slightly smaller right endpoint. Identity (6.18), rather than numerical optimization, proves the bound on the entire first piece.

Proposition 6.3. *For the reciprocal class (6.1), uniformly for $M \leq T^{1/2}$ and $1 \leq H \leq MT^{-\vartheta}$,*

$$|S(H, M, T; F)| \ll_{\varepsilon} HT^{\vartheta+\varepsilon}. \quad (6.20)$$

The estimate is uniform for subintervals of the two dyadic summation ranges and for weights with bounded variation norm.

Proof. The ranges excluded from (6.4) satisfy the claim by Lemma 6.1. In the hard range, one of (5.3) or (5.5) applies. Insert Theorem 1.2 into (5.8), use (5.11), and in Case B compare the four endpoint monomials using (5.13) and (6.8). At N_A , Lemma 6.2 permits the square-root bound. Equations (6.18) and (6.19) prove that its exponent is at most ϑ throughout (6.10). The fixed compact q interval makes all constants uniform. The logarithmic factors in the block length and the finite number of dyadic decompositions are absorbed into T^{ε} after choosing smaller auxiliary exponents. The statement about subintervals and weights follows from the corresponding uniformity of the interface and partial summation. $\square$

7. FROM RECIPROCAL SUMS TO THE DISCREPANCIES

7.1. A finite Fourier approximation with controlled coefficients. We use $\psi(t) = t - [t] - 1/2$, including the value $-1/2$ at integers. Vaaler's finite approximation [12] gives, for an integer $Y \geq 1$,

$$\psi_Y(t) = - \sum_{h=1}^Y \frac{W(h/(Y+1))}{\pi h} \sin(2\pi ht), \quad (7.1)$$

where

$$W(u) = \pi u(1-u) \cot(\pi u) + u \quad (0 < u < 1),$$

with the continuous endpoint values. It satisfies

$$|\psi(t) - \psi_Y(t)| \leq \frac{1}{2(Y+1)} \sum_{|h| \leq Y} \left(1 - \frac{|h|}{Y+1}\right) e(ht). \quad (7.2)$$

The right side is a nonnegative Fejér kernel divided by $2(Y+1)$. At an integer it equals $1/2$, so the convention for ψ at its discontinuities is included. The function W extends smoothly to $[0, 1]$ and has bounded variation there; this also follows by expanding the cotangent at the two endpoints.

Lemma 7.1. *Let F belong to (6.1) and $M \leq T^{1/2}$. For any integer subinterval $I \subset [M, 2M)$,*

$$\left| \sum_{m \in I} \psi \left(\frac{T}{M} F(m/M) \right) \right| \ll_{\varepsilon} T^{\vartheta+\varepsilon}. \quad (7.3)$$

Proof. For $M \leq 2T^{\vartheta}$ the result follows from $|\psi| \leq 1/2$. Otherwise take $Y = \lfloor MT^{-\vartheta} \rfloor$, so $Y \asymp MT^{-\vartheta}$ and $Y \geq 1$. Put $u_m = (T/M)F(m/M)$.

On each dyadic interval $H \leq h < 2H$, the coefficient $W(h/(Y+1))/h$ has supremum plus total variation $O(H^{-1})$. Apply summation by parts in h to the rectangle estimates Proposition 6.3. The corresponding part of the main sum (7.1) is $O_{\varepsilon}(T^{\vartheta+\varepsilon})$.

Summing (7.2) over m gives a constant term $O(M/Y)$ and the real part of a weighted double sum with coefficient

$$\frac{1}{Y+1} \left(1 - \frac{h}{Y+1}\right).$$

On a dyadic interval its variation norm is $O(Y^{-1})$. Thus its contribution is $O_{\varepsilon}((H/Y)T^{\vartheta+\varepsilon})$. Sum the dyadic values of H ; their sum is $O(Y)$. The error is therefore $O(M/Y + T^{\vartheta+\varepsilon})$. Finally $M/Y \ll T^{\vartheta}$, and the logarithm from the main Fourier sum is absorbed into T^{ε} .

This argument bounds the Fourier remainder by weighted double sums. It does not replace their h -sum by the sum of individual absolute values. $\square$

7.2. The divisor problem. Let $U = \lfloor \sqrt{X} \rfloor$. The elementary hyperbola identity gives

$$\sum_{n \leq X} d(n) = 2 \sum_{m \leq U} \lfloor X/m \rfloor - U^2.$$

Using $\lfloor t \rfloor = t - \psi(t) - 1/2$ and

$$\sum_{m \leq U} \frac{1}{m} = \log U + \gamma + \frac{1}{2U} + O(U^{-2}),$$

one obtains

$$\Delta(X) = -2 \sum_{m \leq U} \psi(X/m) + O(1). \quad (7.4)$$

To see that the error is $O(1)$ rather than $O(\sqrt{X})$, put $U = \sqrt{X} + O(1)$ in the displayed expansion. The terms linear in $U - \sqrt{X}$ cancel; the remaining error is $O((U - \sqrt{X})^2 + X/U^2)$.

Decompose $m \leq U$ into dyadic intervals and apply Lemma 7.1 with $T = X$ and $F(t) = 1/t$. The finitely truncated last interval is allowed. This proves the divisor assertion of Theorem 1.1.

7.3. The circle problem and the character modulo four. Let χ_4 be the nonprincipal character modulo four, extended by zero on even integers, and put

$$A(u) = \sum_{1 \leq d \leq u} \chi_4(d).$$

The classical two-squares identity gives, for real $X \geq 1$,

$$\#\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \leq X\} = 1 + 4 \sum_{dm \leq X} \chi_4(d). \quad (7.5)$$

For reference, the coefficient identity is $r_2(n) = 4 \sum_{d|n} \chi_4(d)$ for $n \geq 1$. It follows either from the theta-series identity or from unique factorization in $\mathbb{Z}[i]$; the sum on the right is multiplicative, with local factors $e + 1$ at a prime $1 \pmod{4}$, 1 or 0 according as the exponent is even or odd at a prime $3 \pmod{4}$, and 1 at the prime 2. These are precisely the two-square representation factors after the four units are accounted for.

We derive the needed sawtooth identity so that all shifts and endpoint terms are explicit. With $U = \lfloor \sqrt{X} \rfloor$, the hyperbola decomposition is

$$\sum_{dm \leq X} \chi_4(d) = \sum_{d \leq U} \chi_4(d) \lfloor X/d \rfloor + \sum_{m \leq U} A(X/m) - UA(U). \quad (7.6)$$

The character partial sum has the exact expression

$$\begin{aligned} A(u) &= \left\lfloor \frac{u+3}{4} \right\rfloor - \left\lfloor \frac{u+1}{4} \right\rfloor \\ &= \frac{1}{2} + \psi\left(\frac{u}{4} + \frac{1}{4}\right) - \psi\left(\frac{u}{4} + \frac{3}{4}\right). \end{aligned} \quad (7.7)$$

As a function of the real variable u , $A(u)$ is periodic of period four and has average $1/2$. The primitive of $A(u) - 1/2$ is bounded. Partial summation therefore yields

$$\sum_{d > U} \frac{\chi_4(d)}{d} = -\frac{A(U)}{U} + \int_U^\infty \frac{A(t)}{t^2} dt = \frac{1/2 - A(U)}{U} + O(U^{-2}). \quad (7.8)$$

Since $\sum_{d \geq 1} \chi_4(d)/d = \pi/4$, inserting (7.7) and (7.8) into (7.6) gives

$$\begin{aligned} R(X) &= -4 \sum_{d \leq U} \chi_4(d) \psi(X/d) \\ &\quad + 4 \sum_{m \leq U} \left[\psi\left(\frac{X}{4m} + \frac{1}{4}\right) - \psi\left(\frac{X}{4m} + \frac{3}{4}\right) \right] + O(1). \end{aligned} \quad (7.9)$$

Indeed, the only prospective terms of size U in this substitution are

$$-\frac{X}{U}(1/2 - A(U)) + U(1/2 - A(U)),$$

which are $O(1)$ because $X - U^2 = O(U)$. All other tail terms are $O(X/U^2 + 1) = O(1)$.

In the first sum of (7.9), split the odd integers into $d = 4j + 1$ and $d = 4j + 3$. The terms with $j = 0$ contribute $O(1)$. On a dyadic interval $j \sim M$, the phase is

$$\frac{X}{4j+a} = \frac{X}{M} \frac{1}{4(j/M) + a/M}, \quad a \in \{1, 3\}.$$

Thus $F(t) = \frac{1}{4}(t + a/(4M))^{-1}$ belongs uniformly to the reciprocal class. In the other two sums take

$$F(t) = \frac{1}{4t} + \frac{M\beta}{X}, \quad \beta \in \{1/4, 3/4\}.$$

The constant is placed inside the phase, not treated as a rapidly varying bounded-variation weight in h . Apply Lemma 7.1 with $T = X$ to each dyadic interval. This proves the circle assertion and completes the proof of Theorem 1.1.

8. THE REMAINING LOSS IN THIS ARGUMENT

The amplitude decomposition removes the need to charge the radial energy for cross-ray coherence. It does not eliminate the cross-ray term

$$K^{q-2} L^{(q-1)/2}.$$

At the interior maximum x_0 , it is this term that imposes

$$(q-4)\kappa(x_0) = \lambda(x_0), \quad q = 4.334153950333945\dots$$

Its source is the crossing scale (4.7), where a local density bound and the nonradial arithmetic energy have equal size. Thus the obstruction in the present proof is no longer the all-equal- k family at the finest scale.

There are other potential sources of cancellation that this proof does not use. The original short-block sums are still treated with the absolute values in the quoted arithmetic construction, and the first and second spacing quantities are still separated by a large sieve. Retaining their phases or proving an incompatibility between their extremizing configurations would change a different part of the method. The four-dimensional Poisson geometry and higher reciprocal jets are not required for any theorem of this paper.

Even a square-root first-spacing bound throughout $4 \leq q \leq 9/2$, with no parameter restriction, would give only

$$\theta_{\text{ideal}}(q) = \frac{49q + 66}{4(41q + 44)} \geq \frac{573}{1828} > \frac{5}{16}$$

with the present block lengths and second-spacing factor. This is a limitation of this specified parameter architecture, not an impossibility statement for first spacing or for the circle problem.

APPENDIX A. PARAMETER IDENTITIES AND PROOF DEPENDENCIES

The following identities explain every monomial comparison used in the arithmetic part. Ignoring logarithms only after allowing T^ε , write $H = MT^x$. Then

$$\begin{aligned} \log_T(N_A/M) &= -16x/25 - 49/100, \\ \log_T(R_0/M) &= 8x/25 - 51/200, \\ \log_T K_0 &= -24x/25 - 47/200, \\ \log_T L_0 &= 17x/25 + 51/200, \\ \log_T(K_0 L_0) &= 1/50 - 7x/25. \end{aligned}$$

For a monomial $K^a L^b$ in the q th-power first-spacing bound, its Q/R_0 power after (5.10) and the large sieve is $a + b + 8 - 3q$. Its N power after taking the q th root is

$$-\frac{3}{2} + \frac{11}{17q} + \frac{3a + b}{2q}.$$

For the four monomials these are:

Monomial	Q/R_0 power	N power
$(KL)^{q/2}$	$8 - 2q$	$-1/2 + 11/(17q)$
KL^{2q-5}	$4 - q$	$-1/2 - 6/(17q)$
$K^{q-2}L^{(q-1)/2}$	$(11 - 3q)/2$	$1/4 - 177/(68q)$
$K^{3q/4-2}L^{5q/4-2}$	$4 - q$	$1/4 - 57/(17q)$

The proof has two independent input branches. The analytic branch is the Guth–Maldague theorem (2.8), followed by Lemmas 2.1, 2.2, 3.1 to 3.3 and 4.1 and Proposition 4.2 and Theorem 1.2. The arithmetic branch is the Li–Yang interface (5.8), the admissible lengths (5.4)–(5.6), and the classical estimates in Lemma 6.1. They meet in Proposition 6.3. Vaaler’s finite approximation, the hyperbola identities, and Lemma 7.1 then prove Theorem 1.1.

The interval and sign checks in Section 6 can also be made without derivatives. If ϑ is as in (1.4), then

$$(9396 - 25280\vartheta)^2 - 8832(1963 - 5480\vartheta) = 0.$$

This is precisely the zero-discriminant identity behind (6.18). The lower root gives the maximum on the interval with $632x + 137 < 0$. The upper root would correspond to a different stationary configuration and is not used.

APPENDIX B. A DIRECT CHECK OF THE LOCALIZATION CONVENTIONS

There are three distinct uses of localization, and keeping them separate is important.

First, the common cutoff $\chi(z/P)$ replaces a periodic exponential sum by a Schwartz function. Its Fourier support is fixed before any angular projection is taken. This support is chosen small enough both for the cone neighborhood and for the exact integer equations in Lemma 3.3.

Second, the two angular grids in Lemma 2.1 do not smooth the angular boundary of a Fourier atom. Each atom is kept intact by one sharp angular projection. The assignment is made by k , so the decomposition into same-ray squares continues to be exact after the angular split. The constants in the external theorem are unchanged by the one rotation used for the second class.

Third, each wave-envelope average is a genuinely weighted integral, not a periodic average. Its pair kernel is bounded by Lemma 2.2. Only after weighted Cauchy–Schwarz and summation over the envelope translates do we use the global fourth-moment integral. At that point the remaining weight is exactly $|\chi(z/P)|^4$, and its small Fourier support forces the two integer equations. The approximate square-root equation is discarded only after those two exact equations have been recovered.

These distinctions also apply to complex coefficients. Local cross-ray terms may have either sign; Lemma 4.1 handles that sign. The arithmetic counts are upper bounds for absolute values of coefficient products after the relevant nonnegative global energy has been expanded. They are not assertions that the original oscillatory coefficients are positive.

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